

ELECTROMAGNETIC INDUCTION

Electromagnetic induction

- ✓ Describe the concepts of magnetic flux, magnetic flux density, electromagnetic induction, electromotive force (EMF), Faraday's Law and Lenz's Law. ✓
- ✓ Solve problems involving the magnetic flux in an electric current-carrying loop using $\Phi = BA \cos \theta$.
- ✓ Describe the process of inducing an EMF across a moving conductor in a magnetic field.
- ✗
 - Explain how Lenz's Law is consistent with the principle of conservation of energy.
 - Explain how transformers work in terms of Faraday's Law and electromagnetic induction.
 - Solve problems involving electromagnetic induction using $emf = -\frac{N\Delta(BA_{\perp})}{\Delta t}$, $emf = -N\frac{\Delta\phi}{\Delta t}$, $I_p V_p = I_s V_s$ and $\frac{V_p}{V_s} = \frac{N_p}{N_s}$. ✓
 - Describe the concept of an electromagnetic wave.
 - Explain the relationship between oscillating electric charges and electromagnetic waves.

- Describe the concepts of magnetic flux, magnetic flux density, electromagnetic induction, electromotive force (EMF), Faraday's Law and Lenz's Law.

- * **Magnetic Flux** : A measurement of the total magnetic field that passes through a given area. A measure of the number of magnetic field lines passing through the given area.
 (ϕ) $\leftarrow$ phi
 $[Wb]$
 $\leftarrow$ Weber
- * **Magnetic flux density** : The strength of a magnetic field or the number of magnetic field lines per unit area.
 (B)
 $[Wb/m^2, T]$
- * **Electromagnetic Induction** :
 The production of an electromotive force (EMF) across an electrical conductor due to its dynamic interaction with a magnetic field.

* Electromotive force (EMF)

(EMF) a difference in potential that tends to give rise to an electrical current. Measured in volts (V).

* Faraday's law : A law stating that when the magnetic flux linking a circuit change, an electromotive force (EMF) is induced in the circuit proportional to the rate of change of the flux linkage

* Lenz's Law : A law states that the direction of an induced electric current is such that it produces a current whose magnetic field opposes the change in the circuit or the magnetic field that produces it.

- Solve problems involving the magnetic flux in an electric current-carrying loop using $\phi = BA \cos \theta$.

$$\text{Magnetic field strength} = \text{Flux density}$$

$$1 \text{ T} = 1 \text{ Wb} \cdot \text{m}^{-2}$$

If a single arrow was used to represent 1 line of flux (1 Wb), then 4 arrows flowing through an area of 1 m^2 would be a flux density of $4 \text{ Wb} \cdot \text{m}^{-2}$ or 4 T.

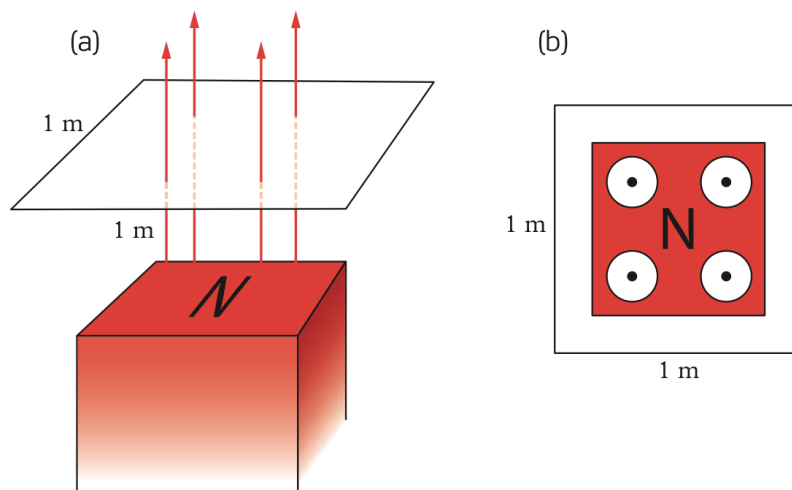


FIGURE 1 (a) Four lines of flux through an area of 1 m^2 represents a magnetic field strength of 1 Wb m^{-2} or 1 T. (b) $B = 4 \text{ T}$ as seen from above with flux lines pointing towards you.

$$\text{Magnetic field strength} = \frac{\text{Magnetic flux}}{\text{Area}}$$

$$B = \frac{\phi}{A}$$

$$\Rightarrow \phi = BA$$

WORKED EXAMPLE 8.1A

Calculate the flux threading a circular loop of diameter 24 cm with the plane of the loop at right angles to a field of strength 5.0 mT.

$$D = 24 \text{ cm} = 0.24 \text{ m}$$

$$r = 0.12 \text{ m}$$

$$A = \pi r^2 = 0.045 \text{ m}^2$$

$$B = 5 \text{ mT} = 5 \times 10^{-3} \text{ T}$$

$$\phi = \underbrace{BA}_{\cos 0^\circ} = 5 \times 10^{-3} \times 0.045 = 2.3 \times 10^{-4} \text{ Wb}$$

Flux at an angle to a field

When the plane of the loop is not at 90° to the field, fewer flux lines will thread the same size loop. When the plane of the loop is parallel to the field for instance, no lines will thread the loop, so the flux in the loop must be zero. A formula is needed that takes the angle into account. The area is a vector quantity and points in the direction perpendicular to its surface (as shown in Figure 3). Because of that the cosine function is used. The product of two vectors such as B and A (the 'dot' product) is given by the product of the two magnitudes multiplied by the cosine of the angle between them.

This is a tricky concept and might take a while to digest. The important part to remember is that the angle a loop makes with a field is the angle between the perpendicular to the loop and the field. That is, when the plane of the loop is at 90° to the field, the perpendicular is at an angle $\theta = 0^\circ$ to the field.

$$\phi = BA \cos \theta \text{ (magnetic flux formula)}$$

where ϕ is the magnetic flux, B is magnetic field strength, A is the area of the loop and θ is the angle that the perpendicular to the plane of the loop makes with a field.

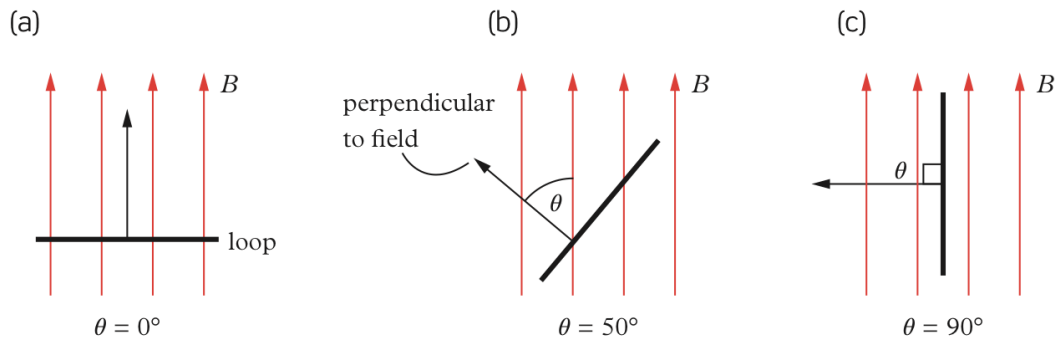


FIGURE 3 A loop rotating in a magnetic field. Note that θ is the angle between the perpendicular to the loop, and the direction of the field.

Consider a loop of 1.0 m^2 with four lines of flux threading it at right angles (Figure 3). This is a magnetic field strength of 4.0 T . The angle θ is the angle perpendicular to the angle the loop (red arrow) makes with the direction of the field (black arrows). In Figure 3a, the loop's perpendicular is parallel to the field, so the angle $\theta = 0^\circ$. In Figure 3b, the loop's perpendicular is at 50° to the field, so $\theta = 50^\circ$. In Figure 3c, the loop's perpendicular is at right angles to the field, so the angle $\theta = 90^\circ$. The calculations for this are shown in Worked example 8.1B.

WORKED EXAMPLE 8.1B

A loop of area 1.0 m^2 is placed in a magnetic field of strength 4.0 T , as shown in Figure 3. Calculate the magnetic flux threading the loop when its perpendicular makes an angle to the field of:

a 50°

b 90°

$$\phi = BA \cos \theta$$

$$\begin{aligned} \text{(a)} \quad \phi &= BA \cos \theta \\ &= 4 \times 1 \times \cos 50^\circ \\ &= 2.6 \text{ Wb.} \end{aligned}$$

$$\text{(b)} \quad \phi = 0 \text{ Wb}$$

Flux inside a solenoid

A solenoid that is carrying an electric current will have a magnetic field inside running from one end to the other, so there will be lines of magnetic flux from one end to the other. To determine the amount of flux, you need to know the magnetic field strength, B , inside the solenoid, and the cross-sectional area, A . Recall from Chapter 7, the formula for the magnitude of the field strength in the solenoid's centre is $B = \mu_0 nI$, where B is the magnitude of the field strength in the solenoid's centre, μ_0 is the permeability constant of $1.26 \times 10^{-6} \text{ T m A}^{-1}$, n is the number of turns per metre of length of the solenoid, and I is the current in ampere.

WORKED EXAMPLE 8.1C

A 25 cm long solenoid of diameter 5.4 cm with 1200 turns of wire carries a current of 1.5 A. Determine the flux inside the solenoid.

$$d = 5.4 \text{ cm} = 0.054 \text{ m}$$

$$L = 25 \text{ cm} = 0.25 \text{ m}$$

$$N = 1200$$

$$I = 1.5 \text{ A}$$

$$\begin{aligned} \text{Number of turns per metre, } n &= \frac{N}{L} \\ &= \frac{1200}{0.25} \\ &= 4800 \text{ turns per metre} \end{aligned}$$

$$\begin{aligned} \text{Cross-sectional Area, } A &= \frac{\pi d^2}{4} \\ &= \frac{\pi \times 0.054^2}{4} \\ &= 0.0023 \text{ m}^2 \end{aligned}$$

$$B = \mu_0 n I = 4\pi \times 10^{-7} \times 4800 \times 1.5$$

$$= 0.0090 \text{ T}$$

$$\Phi = BA \cos \theta$$

$$= 0.0090 \times 0.0023 \times \cos 0^\circ$$

$$= 2.08 \times 10^{-5} \text{ Wb.}$$

- 2 **Calculate** the flux threading a rectangular loop measuring $10 \text{ cm} \times 20 \text{ cm}$ that is placed in a field of strength 4 mT , as shown in Figure 4.

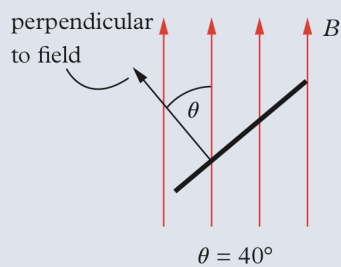


FIGURE 4 Loop at an angle in a field

$$A = 0.1 \times 0.2 = 0.02 \text{ m}^2$$

$$B = 4 \text{ mT} = 4 \times 10^{-3} \text{ T}$$

$$\theta = 40^\circ$$

$$\Phi = BA \cos \theta$$

$$= 4 \times 10^{-3} \times 0.02 \times \cos 40^\circ$$

$$= 0.000061 \text{ Wb}$$

- 3 **Determine** the flux inside a solenoid of diameter 4 cm , given that the solenoid has 800 turns in its length of 15 cm and carries a current of 1.5 A .

$$A = \pi r^2 = \frac{\pi d^2}{4} = \frac{\pi (0.04)^2}{4}$$

$$= 0.00126 \text{ m}^2$$

$$d = 0.04 \text{ m}$$

$$N = 800$$

$$L = 0.15 \text{ m}$$

$$I = 1.5 \text{ A}$$

$$n = \frac{N}{L} = \frac{800}{0.15} = 5333 \text{ turns per metre}$$

$$B = \mu_0 n I = 4\pi \times 10^{-7} \times 5333 \times 1.5 = 0.01 \text{ T}$$

$$\phi = BA \cos \theta = 0.01 \times 0.00126 \times \cos 0^\circ = 0.000051 \text{ Wb.}$$

★ 22 **Determine** which is greater: 15 mT or 0.016 T.

★ 23 **Categorise** the following field strengths from smallest to largest:

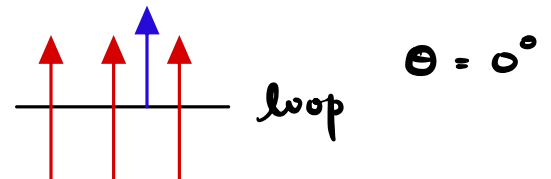
0.011 T, 1.8×10^{-3} T, 15 mT, 16 500 nT

★★ 24 **Determine** the magnitude of magnetic flux in the following situations:

- A 1 cm^2 loop is placed perpendicular to a magnetic field of strength $4.5 \times 10^{-10} \text{ T}$.
- A 0.5 cm^2 loop is placed perpendicular to the magnetic field 5 cm from a wire carrying a steady current of 5 A.
- A 60 mm^2 loop is placed in a magnetic field of strength $6.0 \times 10^{-9} \text{ T}$ such that the plane of the loop makes an angle of 60° to the field lines.
- A loop of diameter 20 cm is placed in magnetic field of strength 5 mT at an angle of 35° between the plane of the loop and the field direction.

$$(a) \quad A = 1 \text{ cm}^2 = 10^{-4} \text{ m}^2$$

$$B = 4.5 \times 10^{-10} \text{ T}$$



$$\phi = BA \cos \theta$$

$$= 4.5 \times 10^{-10} \times 10^{-4} \times 1$$

$$= 4.5 \times 10^{-14} \text{ Wb.}$$

$$(b) \quad A = 0.5 \text{ cm}^2 = 0.5 \times 10^{-4} \text{ m}^2$$

$$r = 5 \text{ cm} = 0.05 \text{ m}$$

$$I = 5 \text{ A}$$

$$B = \frac{\mu_0 I}{2\pi r} = \frac{4\pi \times 10^{-7} \times 5}{2\pi \times 0.05} = 0.00002 \text{ T}$$

$$\begin{aligned} \phi &= BA \cos \theta = 0.00002 \times 0.5 \times 10^{-4} \times \cos 0^\circ \\ &= 1 \times 10^{-9} \text{ Wb.} \end{aligned}$$

(c) $A = 60 \text{ mm}^2 = 60 \times 10^{-6} \text{ m}^2$

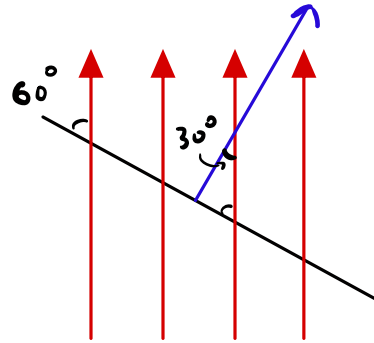
$$B = 6 \times 10^{-9} \text{ T}$$

$$\theta = 30^\circ$$

$$\phi = BA \cos 30^\circ$$

$$= 6 \times 10^{-9} \times 60 \times 10^{-6} \cos 30^\circ$$

$$= 3.1 \times 10^{-13} \text{ Wb.}$$



(d) $d = 20 \text{ cm} = 0.02 \text{ m}$

$$B = 5 \text{ mT} = 5 \times 10^{-3} \text{ T}$$

$$\theta = 55^\circ$$

$$A = \frac{\pi d^2}{4} = \frac{\pi \times 0.02^2}{4} = 3.14 \times 10^{-4} \text{ m}^2$$

$$\begin{aligned} \phi &= BA \cos \theta = 5 \times 10^{-3} \times 3.14 \times 10^{-4} \cos 55^\circ \\ &= 9.01 \times 10^{-7} \text{ Wb.} \end{aligned}$$

→ Electromotive force

- Describe the process of inducing an EMF across a moving conductor in a magnetic field.

The production of an electromotive force (EMF) or voltage across an electrical conductor due to its dynamic interaction with a magnetic field. This process is called electromagnetic induction.

- Solve problems involving electromagnetic induction using

$$emf = -\frac{N\Delta(BA_{\perp})}{\Delta t}, emf = -N\frac{\Delta\phi}{\Delta t}, I_p V_p = I_s V_s \text{ and } \frac{V_p}{V_s} = \frac{N_p}{N_s}.$$

Faraday's law : A law stating that when the magnetic flux linking a circuit change, an electromotive force (EMF) is induced in the circuit proportional to the rate of change of the flux linkage

$$EMF = -\frac{N\Delta(BA_{\perp})}{\Delta t}$$

where N = number of turns

B = magnetic field strength

$A_{\perp}$ = Area that perpendicular to the direction of B .

Δt = time interval

But $\phi = BA \cos \theta$

$$EMF = -N \frac{\Delta \phi}{\Delta t}$$

WORKED EXAMPLE 8.2A

A magnetic field of strength 4.0 mT threads (passes through) a loop of area 1.5 m² at right angles to the plane of the loop (Figure 2). The field strength is suddenly reduced to 1.0 mT in 0.50 s.

- a Calculate the change in flux.
b Calculate the EMF produced.

$$\Delta B = B_{\text{Final}} - B_{\text{Initial}} = 1 \times 10^{-3} - 4 \times 10^{-3} = -3 \times 10^{-3} \text{ T}$$

$$A = 1.5 \text{ m}^2$$

$$\Delta t = 0.5 \text{ s}$$

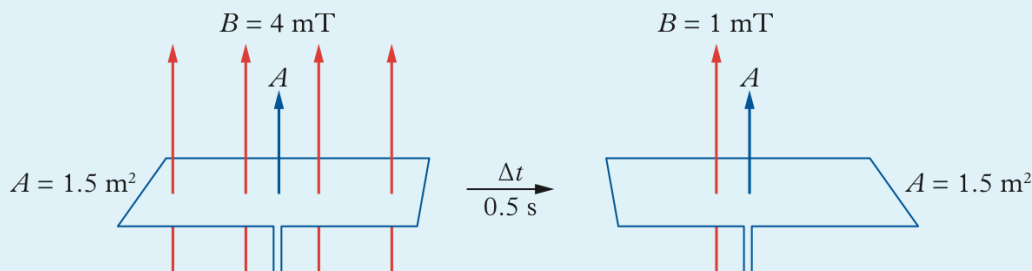


FIGURE 2 Changing flux in stationary loop

$$\begin{aligned} (a) \quad \Delta \phi &= \Delta (BA \cos \theta) \\ &= 1.5 \times (-3 \times 10^{-3}) \times \cos 0^\circ \\ &= -4.5 \times 10^{-3} \text{ Wb.} \end{aligned}$$

$$\begin{aligned} (b) \quad EMF &= -N \frac{\Delta \phi}{\Delta t} = -1 \times \frac{-4.5 \times 10^{-3}}{0.5} \\ &= +9.0 \times 10^{-3} \text{ V} \end{aligned}$$

$$|EMF| = 9 \times 10^{-3} \text{ V}$$

WORKED EXAMPLE 8.2B

Consider the same set-up as in Worked example 8.2A, in which a magnetic field of strength 4.0 mT threads a loop of area 1.5 m² at right angles to the field. The loop is now rotated through 20° in 0.10 s, as shown in Figure 3. Determine the EMF generated in the loop.

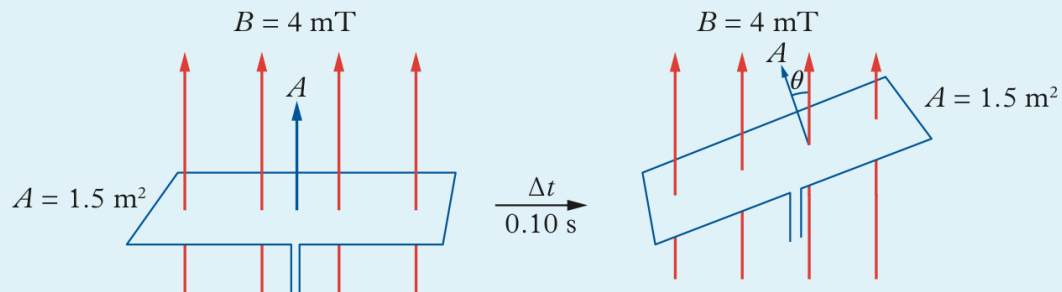


FIGURE 3 Changing the angle θ of loop

$$\theta_{\text{initial}} = 0^\circ$$

$$\theta_{\text{final}} = 20^\circ$$

$$\text{EMF} = - \frac{N \Delta (BA_{\perp})}{\Delta t} = - \frac{N \Delta (BA \cos \theta)}{\Delta t}$$

$\therefore B$ and A remains constant

$$\begin{aligned} \therefore \text{EMF} &= - \frac{NBA \Delta (\cos \theta)}{\Delta t} \\ &= - \frac{NBA (\cos(\theta_{\text{final}}) - \cos(\theta_{\text{initial}}))}{\Delta t} \\ &= - \frac{1 \times 4 \times 10^{-3} \times 1.5 \times (\cos 20^\circ - \cos 0^\circ)}{0.1} \end{aligned}$$

$$|\text{EMF}| = 3.6 \times 10^{-3} \text{ V}$$

WORKED EXAMPLE 8.2C

Consider the same set-up as in Worked example 8.2A, in which a magnetic field of strength 4.0 mT threads (passes through) a loop of area 1.50 m² at right angles to the field. The loop now shrinks in area to 0.60 m² in 0.10 s, as shown in Figure 4. Calculate the EMF induced in the loop.

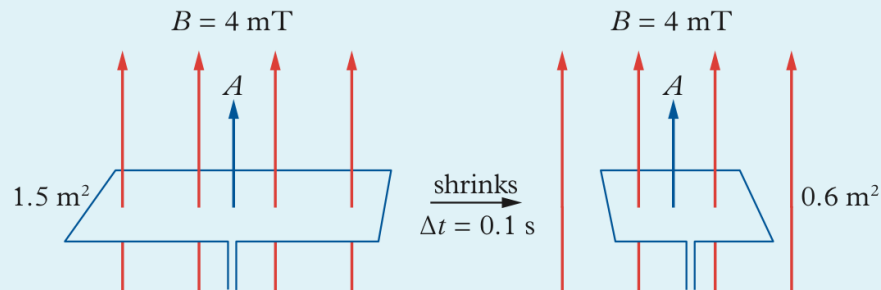


FIGURE 4 Changing the area, A , of the loop

$$\begin{aligned} \text{EMF} &= - \frac{N \Delta (BA \cos \theta)}{\Delta t} \\ &= - \frac{N B \cos \theta \Delta A}{\Delta t} \\ &= - \frac{N B \cos \theta (A_{\text{final}} - A_{\text{initial}})}{\Delta t} \\ &= - \frac{1 \times 4 \times 10^{-3} \times \cos 0^\circ (0.6 - 1.5)}{0.1} \end{aligned}$$

$$|\text{EMF}| = 36 \text{ V}$$

- 5 A square loop of side 25 cm has a magnetic field of strength 2.2 T threading through it. **Calculate** the magnitude of the EMF produced when the field strength falls to 1.0 T in a time of 0.20 s.

$$A = 0.25 \times 0.25 = 0.0625 \text{ m}^2$$

$$\begin{aligned} \text{EMF} &= \frac{-N \Delta(BA \cos \theta)}{\Delta t} = \frac{-NA \cos \theta \Delta B}{\Delta t} \\ &= \frac{-1 \times 0.0625 \times \cos 0^\circ \times (1 - 2.2)}{0.2} \end{aligned}$$

$$|\text{EMF}| = 0.375 \text{ V}$$

- 8 **Determine** the magnitude of the EMF produced in this scenario. A 10-turn coil lies in the plane of this page and a uniform magnetic field of strength 0.90 T is directed into this page. The coil has an area of 0.150 m² and is stretched to have zero area in a time of 0.20 s.

$$B = 0.9 \text{ T}$$

$$N = 10 \quad \theta = 0^\circ$$

$$A_{\text{final}} = 0 \text{ m}^2 \quad A_{\text{initial}} = 0.15 \text{ m}^2$$

$$\Delta t = 0.2 \text{ s}$$

$$\begin{aligned} \text{EMF} &= - \frac{N \Delta(BA \cos \theta)}{\Delta t} = \frac{-NB \cos \theta (A_{\text{final}} - A_{\text{initial}})}{\Delta t} \\ &= \frac{-10 \times 0.9 \times \cos 0^\circ (0 - 0.15)}{0.2} \end{aligned}$$

$$|\text{EMF}| = 6.75 \text{ V}$$

- 9 **Determine** the number of turns in a coil that is pulled between the poles of a magnet so that the magnetic flux decreases from $3.1 \times 10^{-4} \text{ Wb}$ to $1.0 \times 10^{-5} \text{ Wb}$ in 0.020 s . The induced EMF is measured as 0.75 V .

$$\text{EMF} = -N \frac{\Delta\phi}{\Delta t}$$

$$N = - \frac{\text{EMF} \cdot \Delta t}{\Delta\phi} = - \frac{\text{EMF} \cdot \Delta t}{(\phi_{\text{final}} - \phi_{\text{initial}})}$$

$$= - \frac{0.75 \times 0.02}{1 \times 10^{-5} - 3.1 \times 10^{-4}}$$

$$= 50 \text{ turns}$$

Using Lenz's law on a straight conductor

Consider the apparatus depicted simply in Figure 4. It shows a metal rod AB resting on conducting rails connected to a galvanometer, G, on CD. The apparatus is sitting in a magnetic field, directed into the page, as shown. The conductor ABCD forms a loop through which the magnetic field passes (into the page).

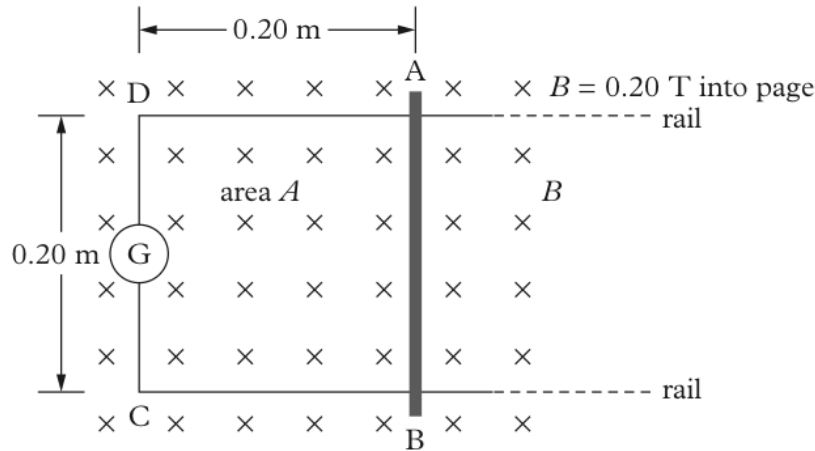


FIGURE 4 A metal rod on conducting rails in a magnetic field forms a loop.

Inducing an EMF by changing the field strength

Imagine the magnetic field strength in Figure 4 changing from a value of 0.20 T to zero in a time of 5.0 ms.

- **Magnitude of the EMF:** The value of the EMF produced is given by Faraday's law:

$$\begin{aligned}
 \text{EMF} &= \frac{n\Delta(BA_{\perp})}{\Delta t} \\
 &= -\frac{nA_{\perp}\Delta B}{\Delta t} \quad (A \text{ is not changing}) \\
 &= -\frac{nA_{\perp} \times (B_f - B_i)}{\Delta t} \\
 &= -\frac{1 \times 0.20 \times 0.20 \times (0 - 0.20)}{5.0 \times 10^{-3}} \\
 |\text{EMF}| &= 1.6 \text{ V}
 \end{aligned}$$

- **Direction of the induced EMF and current:** The direction of the induced current around the loop ABCD can be determined using Lenz's law. The logic is that field strength (B) is decreasing, so the loop wants to oppose this change and keep field strength the same. It does this by generating a field in the same direction as the original field to make up for the losses. The right-hand for loops and solenoids (see Section 7.3, Figure 4b, page 193) shows that the current must flow clockwise from D to A to B to C.

Inducing an EMF by moving the rod

Imagine now, that the rod is moved to the right at a velocity of, say, 5.0 cm^{-1} , with the magnetic field remaining constant at its initial value. The induced EMF from D to C can be calculated.

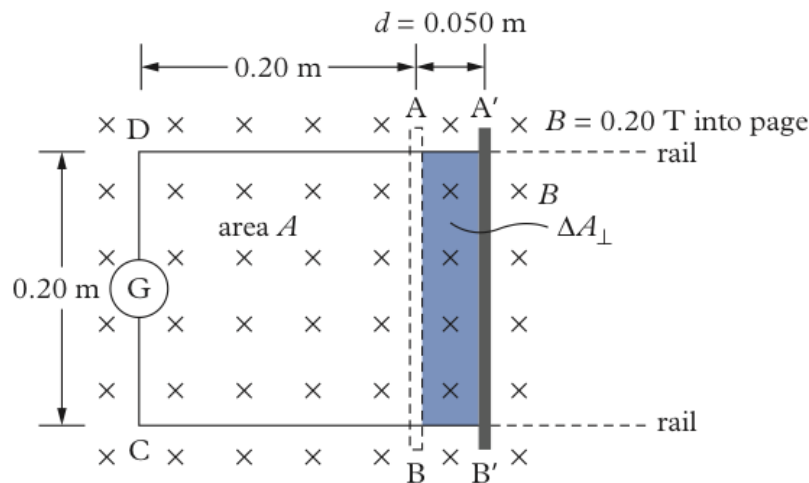


FIGURE 5 The shaded area contains the flux cut by the moving rod.

- **Magnitude of the induced EMF:** In one second of time, the rod cuts (or ‘sweeps’) through 5.0 cm (0.050 m) of the magnetic field (shown in blue as the rod AB moves to new position $A'B'$ in Figure 5). This is an increase in area of $0.050 \times 0.20 = 0.010 \text{ m}^2$, so the change in area (ΔA) is 0.010 m^2 .

$$\begin{aligned} \text{EMF} &= -\frac{n\Delta(BA_{\perp})}{\Delta t} \\ &= -\frac{nB\Delta A_{\perp}}{\Delta t} \quad (B \text{ is not changing}) \\ &= -\frac{1 \text{ turn} \times 0.20 \text{ T} \times 0.010 \text{ m}^2}{1.0 \text{ s}} \\ |\text{EMF}| &= 2.0 \times 10^{-3} \text{ V (2 sf)} \end{aligned}$$

- **Direction of the induced EMF and current:** To determine the direction, use Lenz’s law. When the rod slides to the right, the amount of flux enclosed by the loop increases (Figure 5). The loop opposes this change and generates a field that is in the opposite direction to the existing field to get it back to the original value. Using the right-hand rule for loops and solenoids (see Section 7.3, Figure 4b, page 193), an anticlockwise current (from B' to A') is induced to achieve this.

Alternatively, you can use Fleming’s right-hand generator rule. With index finger (magnetic field, B) pointing into the page, thumb (force, F) pushes wire (AB) to the right, and your middle finger (current, I) points up the page (from B' to A').